



# **CITYFARM**

**MURRAY GROVES, NIC HARROD, LIAM CARTA, FINN COOPER  
AND JACQUE DE BOEHLER**

# STAKEHOLDERS AND INITIAL BRIEF

## Brief

- Log livestock data (birth, vaccines, enclosure)
- Keep track of tasks and events for each animal/enclosure
- Simplify existing system of messy excel spreadsheet and outlook calendar
- Move animals from one farm to another

## Stakeholders

- Bill Lambert (Livestock Manager at Windmill Hill City Farm)
- Volunteers at WHCF
- Steve Sayers (WHCF CEO)

# USER STORIES

## **Livestock Manager**

I need to be able to keep track of the many tasks and animals I must deal with daily. I want everything in one easy-to-access place so I can share it with our volunteers.

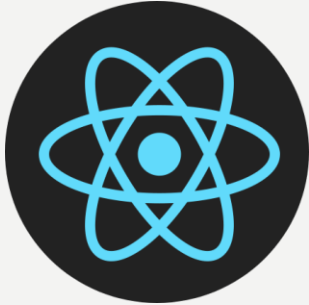
## **Volunteer**

I'm not super experienced at the work we do, and I am not familiar with the complicated excel sheet setup they use now. I need a simple system where I can easily understand what tasks I need to do - and update them as I complete them.

## **CEO**

I sometimes need to be able to see an overview of all our animals and what events are coming up so I can help organise the equipment and people that are needed. Right now, I must ask the Livestock Manager because I don't understand his Excel spreadsheet system. Having a website with all the information in would make this much quicker.

# TECH STACK



## React / Typescript

- We use MUI components to speed up development
- Wrote a library to allow us to leverage Typescript and cache objects



## Spring Boot

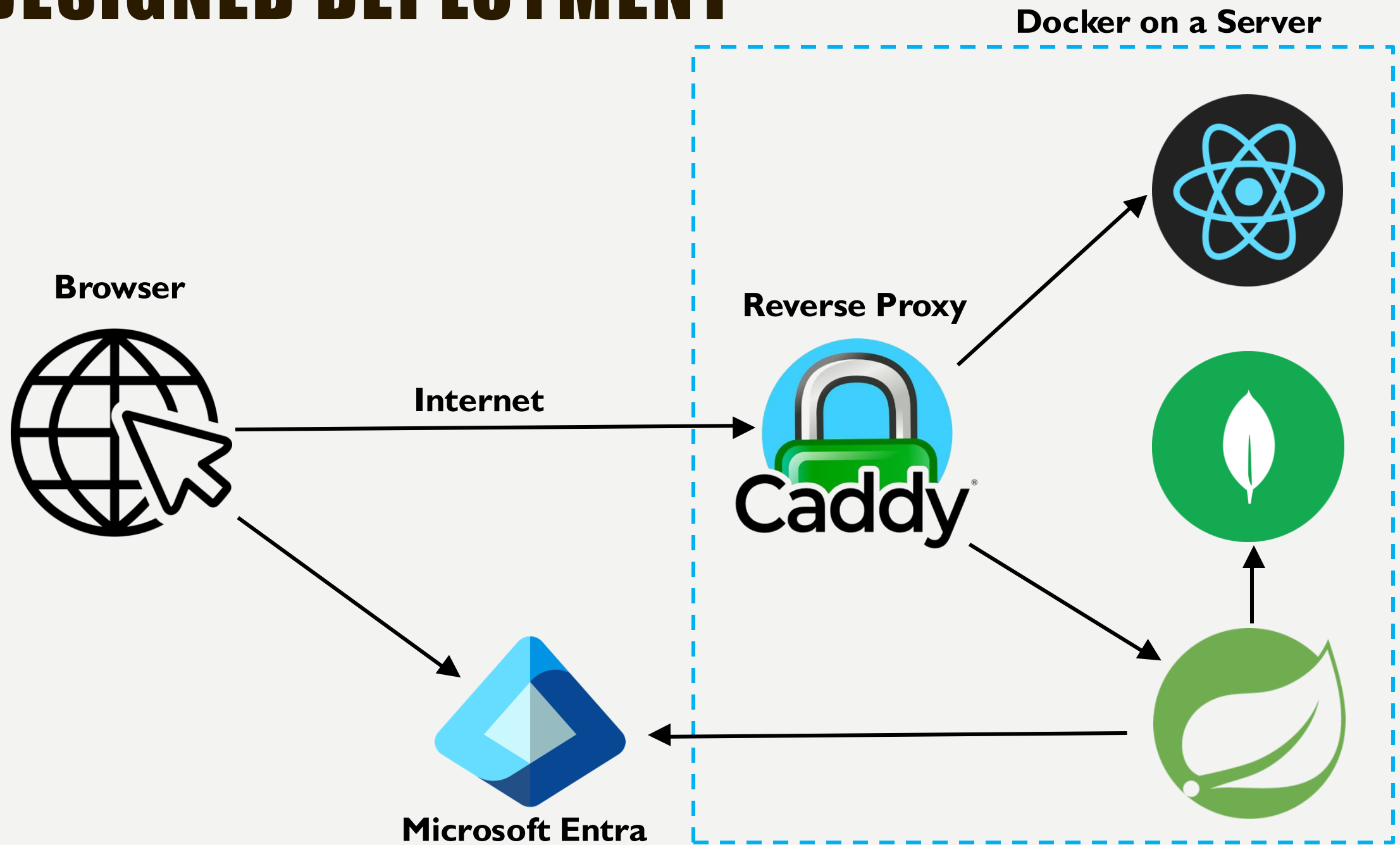
- Relatively simple
- Performs validation & transformation of data
- Powerful integration with MongoDB, ability to define custom serializers



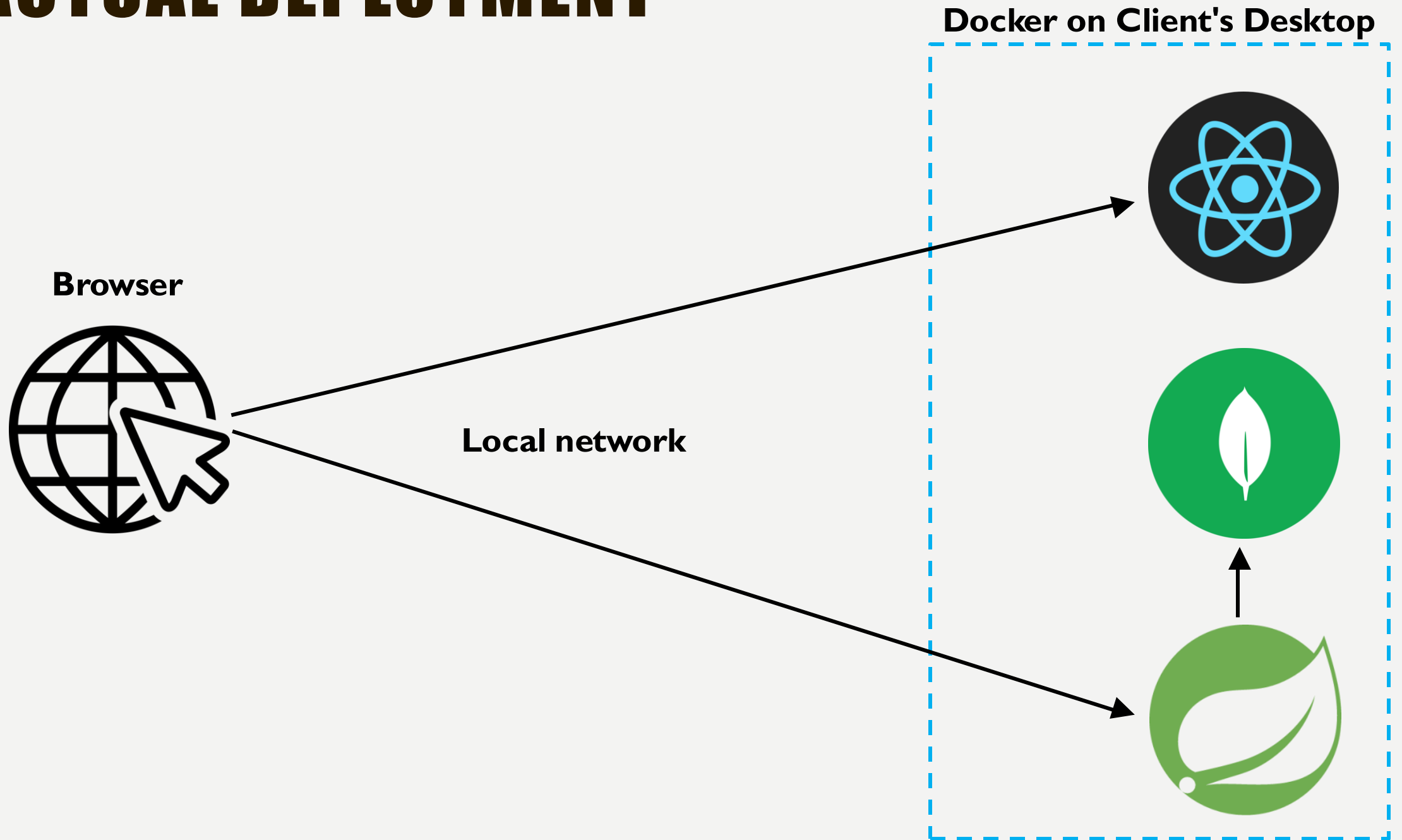
## MongoDB

- Stores all our data
- NoSQL format allows for rapid development and ability to change database schemas regularly

# DESIGNED DEPLOYMENT



# ACTUAL DEPLOYMENT



# BEFORE/AFTER MATERIAL UI

Home

Calendar

Livestock

Enclosures

Livestock

Search

Q Clear Name

Windmill Hill

Hartcliffe

St Werburghs

| Name | Type | Father       | Mother       | Sex    |
|------|------|--------------|--------------|--------|
| Jeff | COW  | Unregistered | Unregistered | Female |
| Jeff | COW  | Unregistered | Unregistered | Female |

name

Chicken

Unregistered

Unregistered

Male

Add

Home

Calendar

Livestock

Enclosures

Animal Types

?

Livestock

Search

ALL

WINDMILL HILL

HARTCLIFFE

ST WERBURGHs

| <input type="checkbox"/> Name     | Type | Father  | Mother       | Sex  |  |
|-----------------------------------|------|---------|--------------|------|--|
| <input type="checkbox"/> Demeter  | Cow  | Alanna  | Unregistered | Male |  |
| <input type="checkbox"/> Gloriane | Cow  | Demeter | Unregistered | Male |  |

CLEAR FILTER

Create Animal

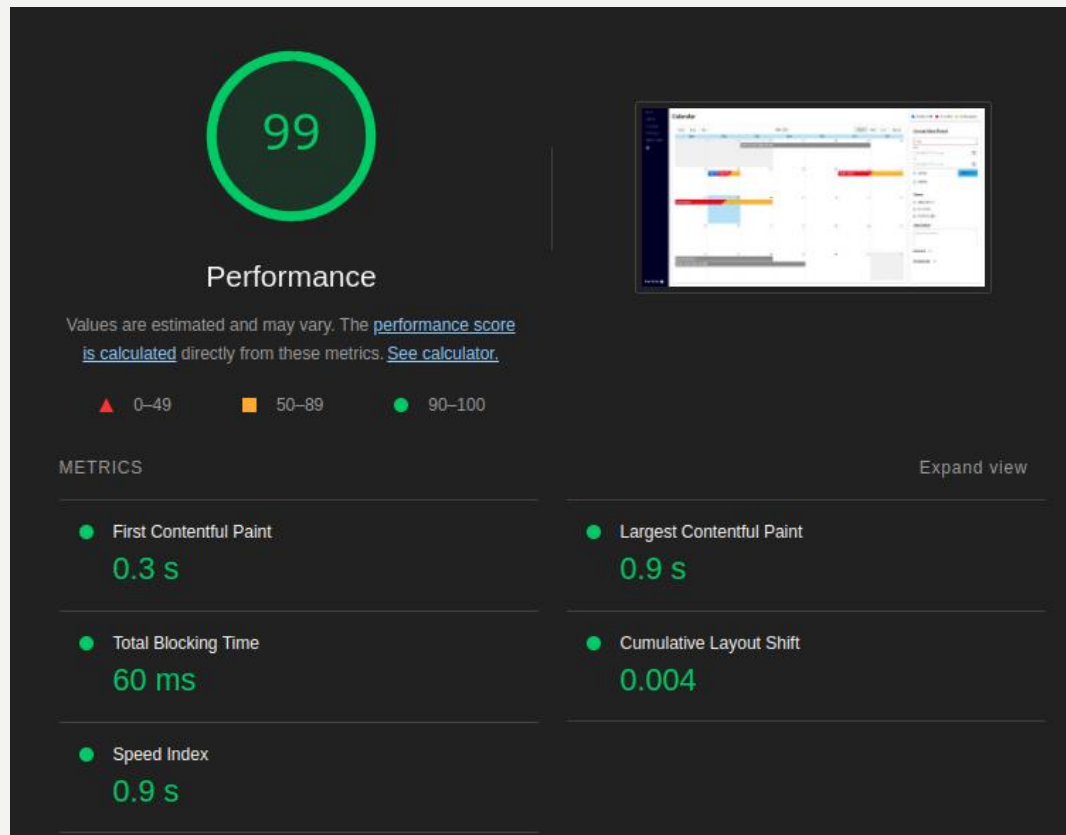
CREATE +

DISCARD

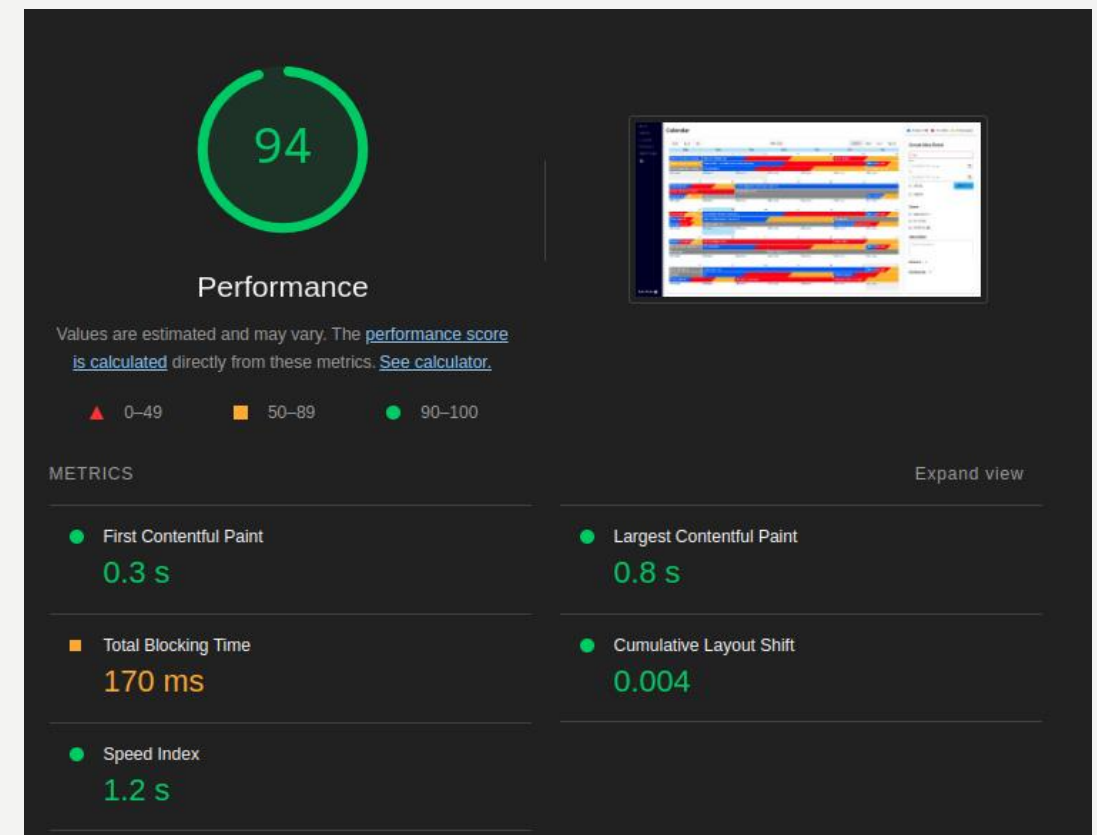
| Name              | Type              | Father            | Mother            | Sex              | Farm            |
|-------------------|-------------------|-------------------|-------------------|------------------|-----------------|
| <div>Name *</div> | <div>Type *</div> | <div>Father</div> | <div>Mother</div> | <div>Sex *</div> | <div>Farm</div> |

# STRESS TESTING

## 20 Events/Animals



## 1 000 Events/Animals

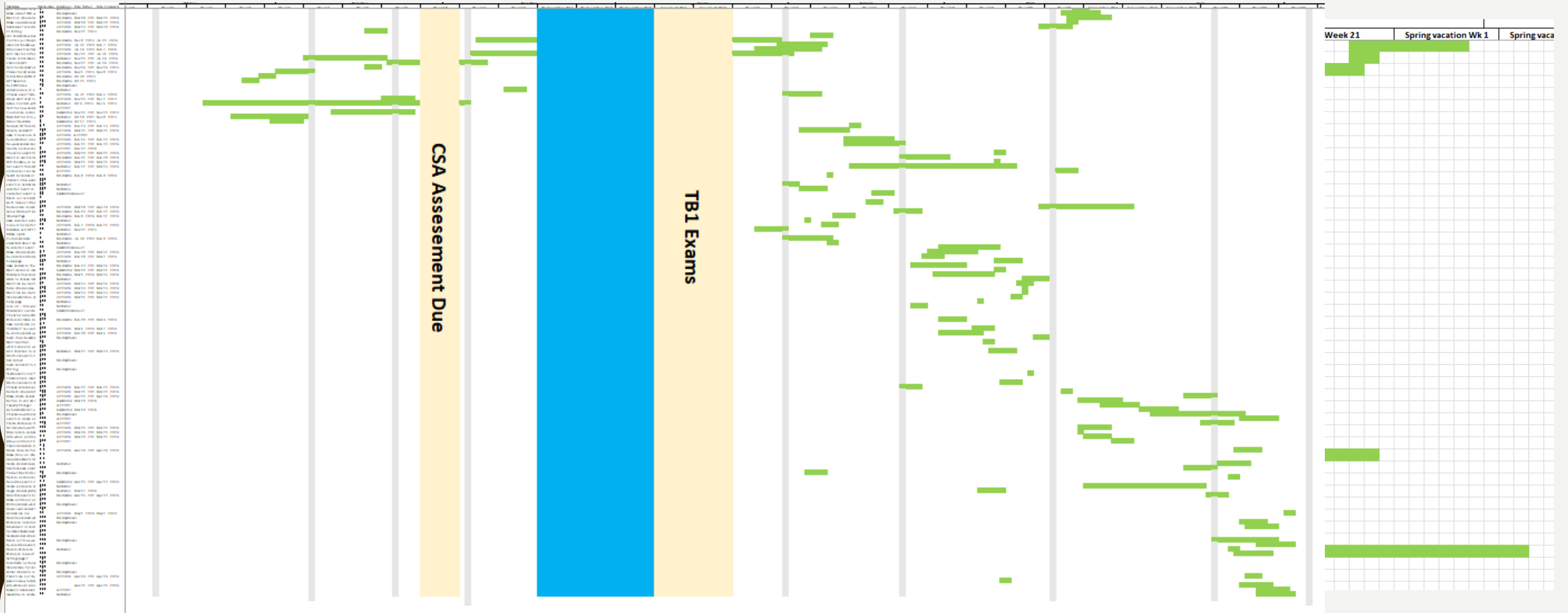


**DEMO**



[cityfarm.murraygrove.es](http://cityfarm.murraygrove.es)

# GANTT CHART AND ANALYSIS OF WORK



Key:

- Yellow – Assessment Period
- Blue – Winter Break
- Gray bars – Client Meeting

# PROCESS

- Very flexible development process
- Each of us pick and choose from issues on the Kanban
- Meet every Wednesday to consolidate work
- Aim to have PRs ready to merge for our weekly meetings
- Communicate over WhatsApp as well as on GitHub for faster responses

# ISSUES WE FACED IN TB1

- Not using the Kanban board to sync work up efficiently meaning people were doing the same work simultaneously
- Made sure people only start work on something if it's available on the kanban, and always mark as in-progress
- Confusion on our tech stack choices
- Briefed whole team on what we were using and where
- Realising we designed the backend in a way that wouldn't allow the client to create new types of animal
- Re-designed it so we can re-implement it in TB2

# CLIENT FEEDBACK - TB1

- Different colors for identifying events based on relevant farms
- Shared calendar between all farms
- Attaching animals/enclosures to events
- Different tabs for different farms on each page
- Events with checkboxes to signify completeness

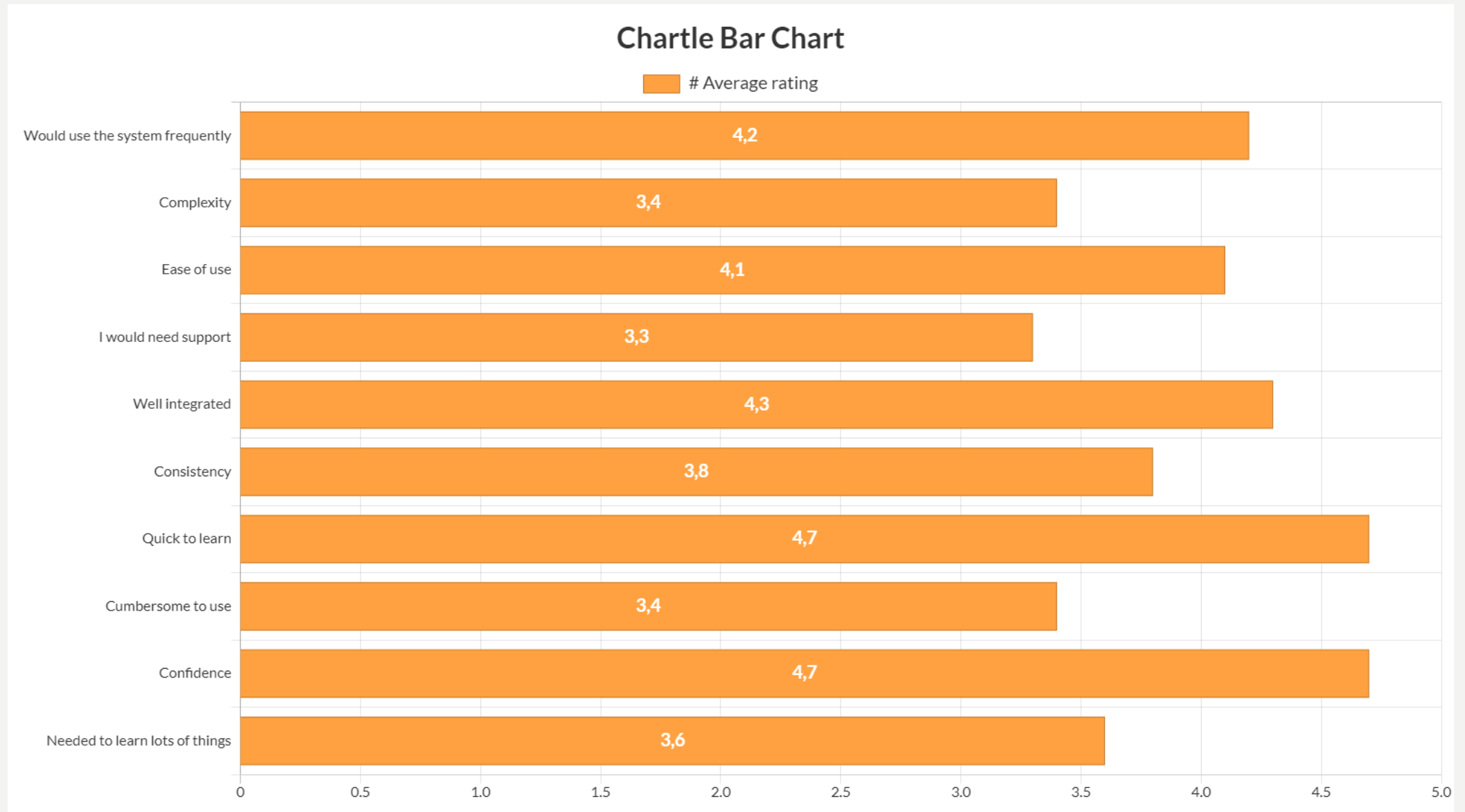
# TB2 PLAN - COMPLETED

- Re-work the backend to allow users to create new animal types and attributes 
- Fully integrate frontend with backend capabilities 
- Frontend quality of life changes such as fields referencing other animals being links to those animals 
- Login system to integrate with their existing Microsoft accounts 
- Support for multiple farms 
- Events with checkboxes to signify completeness 

# CLIENT FEEDBACK - TB2

- Sex isn't male/female, it's male/female/castrated - meant changing from a bool on the backend
- Our client wasn't as responsive as we would have liked since he was very busy and was not very tech savvy

# TESTING DAY RESULTS - NORMALISED



# TESTING DAY – WRITTEN FEEDBACK

*For a farmer or livestock owner it seems quite useful, bit difficult for me as I needed to be told the different fields, but apart from that it was quite good*

*It's not intuitively clear that hovering over an animal name gives more information about it, giving it a different style (like a hyperlink compared to standard text) would make it clearer  
Schema seems like a weird name for a farmer using a template, maybe that should be a word more clearly related to the subject matter.*

*you should be able to see children as well as parents!!!!!!*

Our main takeaway from the first two pieces of feedback was that there was an accessibility issue, which we also saw was evident with the day and telling people how to use it, this led to us creating a home and help page as well as an icon to help with each individual page.

# TESTING DAY RESPONSE

Home  
Calendar  
Livestock  
Enclosures  
Animal Types

## Animal Types

### Create New Animal Type

Name \*

CREATE +

Data Type

This displays a page to create custom animal types, in case any new animals are to be added, once the type name is filled out, any number of properties can be added that pertain to that animal. Once an animal type is added, the creation of animals in the livestock table will be updated to include that.

### Existing Animal Types

Search

Home

Calendar

Livestock

Enclosures

Animal Types



## Help

On the left is the side menu, this contains various ways to manage livestock across the three farms. Below are instructions and help to navigate your way around the application, click the headers to reveal their content.

### Calendar

### Animal Table

This table displays all the animals, each one has 5 main attributes, father, mother, name, type and sex. This can be searched. This table is clickable and will take you to a single animal view page that will display more information about the animal, including move an animal's fields, select it on the grid and then press the move farm buttons below the table. The add animal button is defined in the animal types page and all the fields can be assigned.

### Enclosures

Every enclosure has animals assigned to it. Click on an enclosure to expand it, you can edit them by clicking the edit button.

### Animal Types

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## Murray

Sex: Male

Species: Goat

Father: Unregistered

Mother: Unregistered

Farm: St Werburghs

Notes: Murray is a real one.

Children:

Ricky

Bobby

MOVE TO WH

MOVE TO HC

MOVE TO SW

# ISSUES FACED IN TB2

- Re-working of livestock system to support custom animals took a while
- Ran out of Azure credits
- Ran continuous development on my home server instead
- Some communication issues meant some PRs took longer than they should've to complete and merge
- Made sure everyone checked the WhatsApp group more frequently – used pings to notify reviewers that they have been requested
- Lots of open issues meant it was hard to prioritise – some were also vague
- Combed over whole kanban and added a new "Priority" column – asked issue opener to add more details where necessary

# CLIENT FEEDBACK FROM BETA

- Struggled with login, sometimes didn't let him into the system at all
- Liked how clean everything was visually including the colour coordination in accordance with the different farms' colours
- Would be nice to have it look good on mobile platforms but not necessary

# FUTURE ASPIRATIONS

- Automatic scheduling of events when modifying attributes
- Enable support for more farms
- Synchronising with Outlook Calendar
- Ability to upload pictures of livestock to their 'notes' section in order to more easily recognise them
- Functionality to more intuitively move livestock between farms / enclosures

# QUESTIONS?

